Tables

1

a

x	5	6	6.9	6.99
$f(x)$	0.5	1	10	100

b

x	9	8	7.1	7.01
$f(x)$	-0.5	-1	-10	-100

c The limit does not exist.

2

a

x	1.9	1.99	1.999	2.001	2.01	2.1
$f(x)$	19.05	20.8005	20.980 005	21.020 005	21.2005	23.05

b $\lim_{x \to 2} \left(5x^2 + 1\right) = 21$

3

a

x	-0.1	-0.01	-0.001	0.001	0.01	0.1
$f(x)$	1.0448	1.0049	1.0005	0.9995	0.9950	0.9453

b $\lim_{x \to 0} \left(\dfrac{2 - x}{x^2 + 2}\right) = 1$

4

a

x	3.9	3.99	3.999	4.001	4.01	4.1
$f\left(x\right)$	3.9	3.99	3.999	4.001	4.01	4.1

b $\lim_{x \to 4} \left(\dfrac{x^2 - 4x}{x - 4}\right) = 4$

5

a

x	-1.1	-1.01	-1.001	-0.999	-0.99	-0.9
$f\left(x\right)$	4.31	4.0301	4.003 001	3.997 001	3.9701	3.71

b $\lim_{x \to -1} f\left(x\right) = 4$

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6

a

x	4.9	4.99	4.999	5.001	5.01	5.1
$f(x)$	-62.14	-623.38	-6235.84	6236.29	623.83	62.58

b No, because the function is not approaching the same value on either side of $x = 5$.

7

a

x	-5.1	-5.01	-5.001	-5	-4.999	-4.99	-4.9
$\frac{x^2+4}{x+5}$	-300.1	-2910.01	-29010	$-$	28990	2890.01	280.01

b No, because the function is not approaching the same value on either side of $x = -5$.

8

a

x	2.9	2.99	2.999	3	3.001	3.01	3.1
$\left(\frac{e^{x-3}+x-4}{x-3}\right)$	$1.951\,63$	$1.995\,02$	1.9995	$-$	2.0005	$2.005\,02$	$2.051\,71$

b Yes, because the function is approaching the same value of 2 on either side of $x = 3$.

Graphs

9

a $y\text{-value} = 2$

b $y\text{-value} = -1$

c No, because $g(x)$ approaches 2 on the left of $x = -2$, and $g(x)$ approaches -1 on the right of $x = -2$.

10

a $y\text{-value} = -2$

b $y\text{-value} = -2$

c The limit of $\frac{x^2-1}{x+1}$ as x approaches -1 is -2.

d $\lim_{x \to -1} \left(\frac{x^2-1}{x+1}\right) = -2$

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11 No, because the function approaches different values on either side of $x = 3$.

12 No, because the function approaches different values on either side of $x = 0$.

13 Yes, because of either side of $x = 5$, the function approaches infinity as $x \to 5$.

14 Yes, because on either side of $x = 0$ the function approaches 2.

15 $\lim_{x \to 3} f(x) = 1$

16 $\lim_{x \to 0} f(x) = 0$

17 $\lim_{x \to 3} f(x) = 5$

18 $\lim_{x \to 3} f(x) = 4$

19 $\lim_{x \to 2} f(x) = 1$

20

a $\lim_{x \to -4^+} \left(\frac{1}{x+4} \right) = \infty$

b $\lim_{x \to -4^-} \left(\frac{1}{x+4} \right) = -\infty$

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